Ogbonna Arinze

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SUMMARY

Frontend Engineer with 8 years of experience building interactive, data-driven web applications using React and TypeScript across healthcare, fintech, and civic tech. Specialized in data visualization, having built charting dashboards with react-chartjs and Recharts (22 interactive chart types), real-time streaming interfaces, and performant UI component systems with Ant Design, Radix UI, and Tailwind CSS. Passionate about translating complex data into clear, interactive visual representations with a strong foundation in cross-browser compatibility, component architecture, and performance optimization.

CORE COMPETENCIES

- **Data Visualization:** Recharts (22 chart types), react-chartjs, interactive dashboards, real-time data streaming, and chart-driven UI components.
- **Frontend Development:** React, TypeScript, Next.js, Vue.js, Redux, Zustand, React Query, and modern JavaScript (ESNext).
- **UI Component Frameworks:** Ant Design, Radix UI, Tailwind CSS, custom design systems, Storybook, and cross-browser compatibility.
- **Performance & Quality:** Lighthouse, React Profiler, code splitting, bundle analysis, Playwright, Jest, Cypress, and CI-driven QA.
- **Real-time & Streaming:** EventSource API, SSE streaming, RxJS, WebSockets, and real-time data pipelines.
- **Tools & Infrastructure:** GitHub Actions, Docker, Webpack/Vite/Rollup, Nx monorepos, AWS, and CI/CD pipelines.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, improving workflows for clinical site integrations and ensuring **cross-browser compatibility** across Chrome environments.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for efficient data fetching and request caching, refactoring legacy class components to **TypeScript** functional components and streamlining data flow across the application.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using OpenAI GPT models, **LangChain**, Vercel AI SDK, and Chroma vector database, enabling users to query their data in real time via a conversational interface.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with **TypeScript** and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Increased release reliability by integrating **Playwright** end-to-end tests into the **CI/CD pipeline**, automating reporting with Allure Reports on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built **dynamic data visualization dashboards** using **React** and **react-chartjs**, rendering complex financial datasets into interactive charts for transaction monitoring, settlement tracking, and reconciliation reporting, with **Redux Toolkit** and **Ant Design** UI.
- Played a pivotal role in a team of 3 frontend engineers to build a high-volume bank **Settlement Solution**, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Adopted the native browser **EventSource API** for **real-time data streaming**, replacing conventional socket.io methods for faster data updates and a more responsive interface displaying live transaction feeds.
- Developed a real-time account reconciliation solution, boosting the reconciliation of daily transactions by **60%** through innovative transaction processing and data management strategies.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface across the application.
- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable and well-documented code.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring **22 interactive chart types** auto-generated from AI responses, including bar charts, line charts, pie charts, area charts, and treemaps, with dark mode and responsive mobile layouts.
- Architected the entire platform as an **Nx monorepo** managing **6 applications**: NestJS API backend, Next.js citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, and conversation summarization for responsive, interactive user experiences.
- Built the admin dashboard with user management, usage analytics visualizations, ingestion tracking, and system banner management capabilities.
- Configured CI/CD with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles and health checks.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*